Nabila Alia Zahra	**Rangkuman Minggu 12** **ITU - T**	Selasa, 22 November 2022
		16TNK5043 (Aplikasi Mikrokontroler)
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ITU - T

ITU (The International Telecomunication) → badan dari United Nations

ITU – T Y.2060 (ttg IoT)

Frameworks and functional architecture models

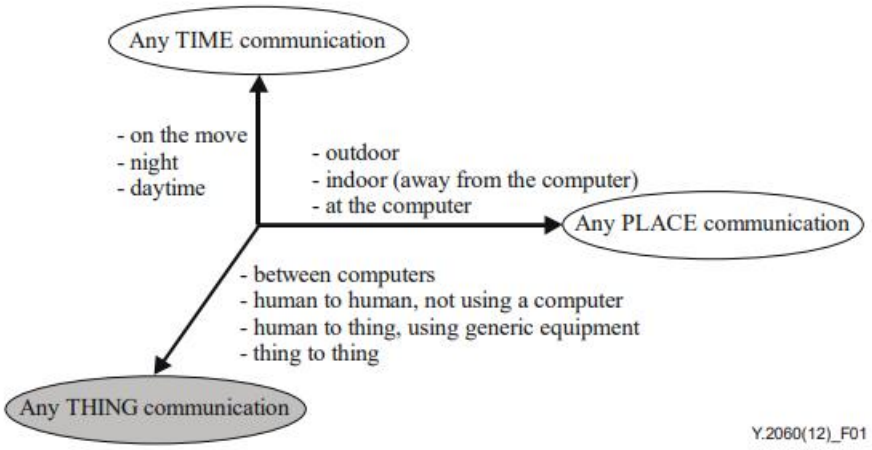
Devices : memiliki kemampuan untuk dapat berkomunikasi, bisa membaca data sensor, mengolah, menyimpan, menampilkan

IoT : A global infrastructure for the information society, enabling advanced services by interconnecting (physical and virtual) things based on existing and evolving interoperable information and communication technologies.

Interkoneksi : fisik dan virtual

*noted : keamanan dan privasi

Data capture, memproses data dan mengirimkannya



Physical thing : represented in the information world via one or more virtual things (mapping), but a virtual thing can also exist without any associated physical thing.

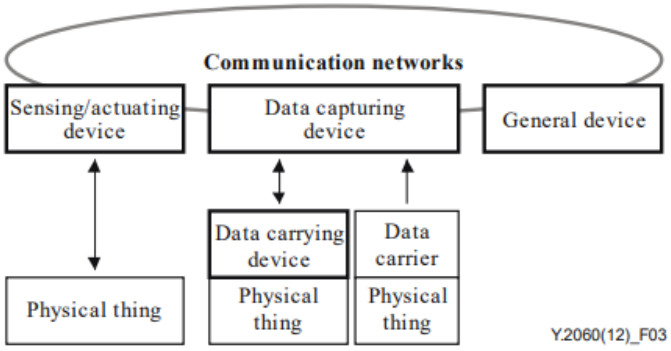


Figure 3 – Types of devices and their relationship with physical things

Virtual word :

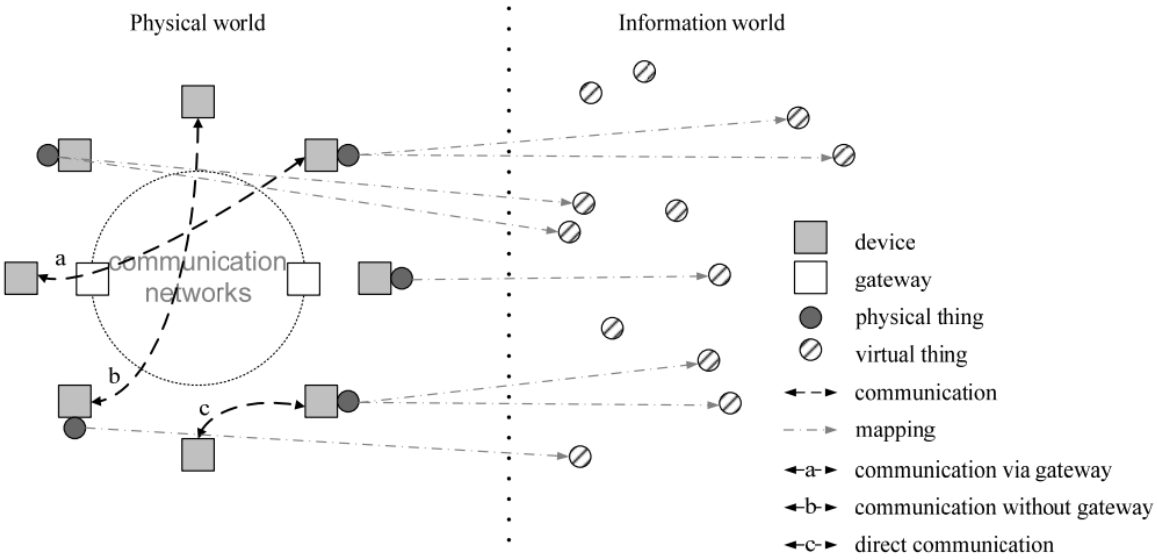


Figure 2 – Technical overview of the IoT

Direct communication : tanpa perlu adanya jaringan untuk berkomunikasi

Karakteristik fundamental

- Interconnectivity: With regard to the IoT, anything can be interconnected with the global information and communication infrastructure.
- Things-related services: The IoT is capable of providing thing-related services within the constraints of things, such as privacy protection and semantic consistency between physical things and their associated virtual things. In order to provide thing-related services within the constraints of things, both the technologies in physical world and information world will change.

- Heterogeneity: The devices in the IoT are heterogeneous as based on different hardware platforms and networks. They can interact with other devices or service platforms through different networks.
- Dynamic changes: The state of devices change dynamically, e.g., sleeping and waking up, connected and/or disconnected as well as the context of devices including location and speed. Moreover, the number of devices can change dynamically.
- Enormous scale: The number of devices that need to be managed and that communicate with each other will be at least an order of magnitude larger than the devices connected to the current Internet. The ratio of communication triggered by devices as compared to communication triggered by humans will noticeably shift towards device-triggered communication. Even more critical will be the management of the data generated and their interpretation for application purposes. This relates to semantics of data, as well as efficient data handling.

Model Referensi IoT (4 Layer) :

- Application layer
- Service support and application support layer
- Network layer
- Device layer.

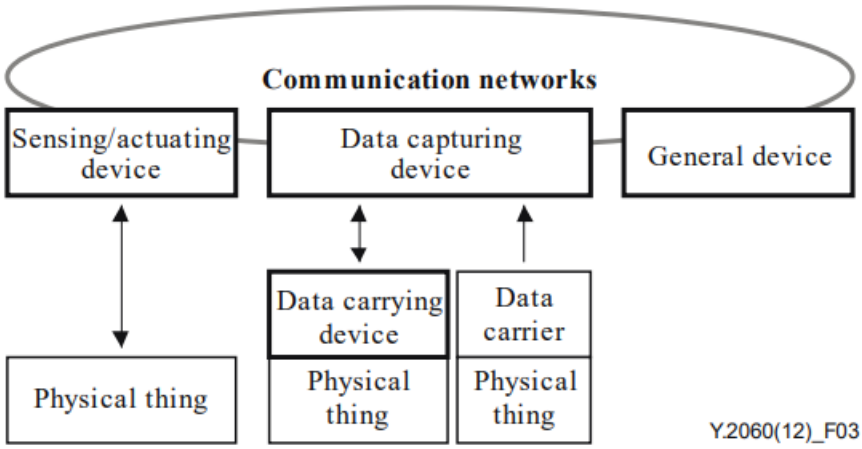
Device : Arduino/system microkontroller

Physical thing : sensor/actuator

Virtual thing : temperature disekitar sensor temperature

Communication network : jaringan internet, jaringan private LAN, Bluetooth

Gateway : yg menghubungkan perangkat komunikasi dengan jaringan komunikasi ➔ wifi router



Capture → simpan → olah → dikirimkan

Minimum requirement IoT:

- Data carrying
- Data capturing
- Sensoring & actuating
-

Fundamentals Characteristics :

- Interconnectivity
- Things related services
- Heterogeneity
- Synamic changes
- Enormous scale

High level requirement :

IoT reference model :

1. Application layer
2. service support and application support layer
3. network layer
4. device layer

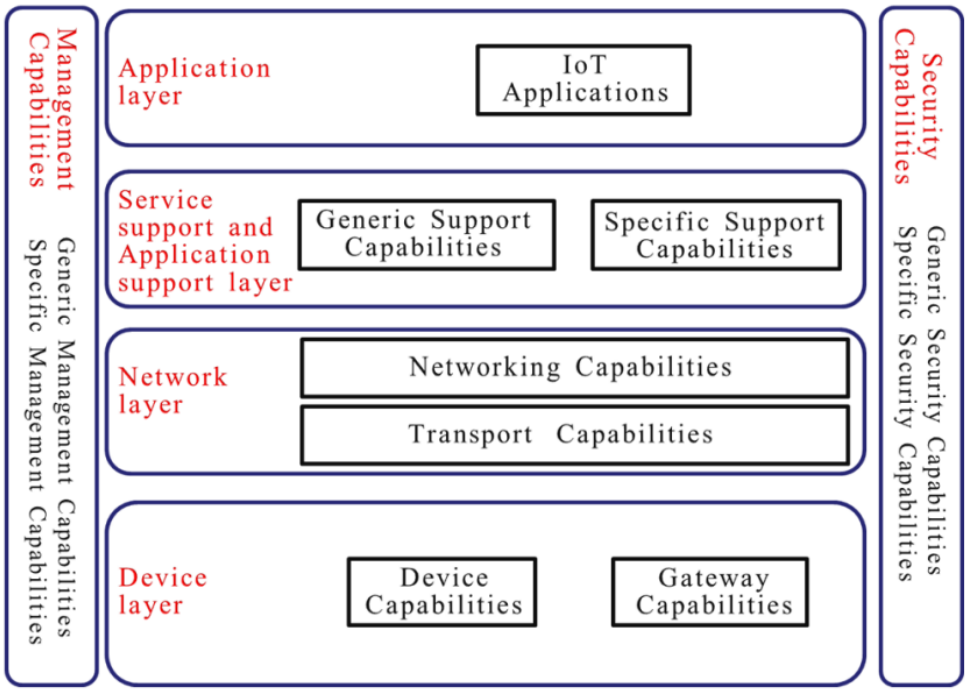


Figure 4 – IoT reference model

1. Membuat program Atmega untuk menampilkan huruf A, b, dan C. ketika menampilkan huruf A dan C tampilan tidak dapat diinterupsi. Ketika menampilkan huruf b tampilan dapat dinterrupsi berulang kali melalui int 1 dan akan menampilkan kata “Halo” selama ½ detik. Huruf A, b, dan C tampil bergantian masing-masing selama 4 detik

```
 .INCLUDE "M8535DEF.INC"
.EQU TIMER_1= 0XFFF0

.ORG 0
RJMP MAIN

.ORG 0X02
RJMP INT_1



MAIN:
        LDI R16,LOW(RAMEND)
        OUT SPL,R16
        LDI R16,HIGH(RAMEND)
        OUT SPH,R16
;-----------------------------------------------------
        LDI R16,0XFF            ;PORT B,C SBG OUTPUT
        OUT DDRB,R16
        OUT DDRC,R16

        LDI R20,0X80            ;ENABLE INT 1
        OUT GICR,R20

        LDI R20,0X08            ;AKTIF KETIKA FALLING
EDGE
        OUT MCUCR,R20

        LDI R28,0X60
        LDI R29,0

        LDI ZL,LOW(2*POSISI)    ;INISIALISASI POINTER Z
        LDI ZH,HIGH(2*POSISI)
        LDI R16,8

;-----------------------------------------------------------
L1:     LPM R19,Z+
        ST Y+,R19
        DEC R16
        BRNE L1

U:      CLI
        LDI R18,0XFE
        OUT PORTB,R18
        LDI R18,0XC6
        OUT PORTC,R18
```

```
                LDI R18,16

KE11:  RCALL DELAY1
                DEC R18
                BRNE KE11

                LDI R18,0X88
                OUT PORTC,R18
                LDI R18,16

KE22:  RCALL DELAY1
                DEC R18
                BRNE KE22

                LDI R16,0X80
                OUT GIFR,R16

                LDI R22,16
SANA1:          LDI R23,0X80
                OUT GICR,R23
                SEI

                LDI R24,0XFE
                OUT PORTB,R24
                LDI R24,0X83
                OUT PORTC,R24

                RCALL DELAY1
                DEC R22
                BRNE SANA1

                RJMP U

INT_1:LDI R27,30
                LDI ZL,LOW(2*POLA)
                LDI ZH,HIGH(2*POLA)
                MOV R20,R30
                RCALL DISPLAY
                RETI

DISPLAY:
                CLI
                MOV R17,R27
KAU1:MOV R30,R20
                LDI R28,0X60
                LDI R25,8
KAU2:LD R0,Y+
                OUT PORTB,R0
                LPM R19,Z+
                OUT PORTC,R19
```

```
                RCALL DELAY
                DEC R25
                BRNE KAU2

                DEC R17
                BRNE KAU1
                RET

DELAY:      CLI
        ;DISABLE GLOBAL INTERUPT KETIKA BERURUSAN DGN
WAKTU
                PUSH R16
                PUSH R17
                LDI R16,4
                OUT TIMSK,R16
                LDI R16, HIGH(TIMER_1)
                OUT TCNT1H,R16
                LDI R16, LOW(TIMER_1)
                OUT TCNT1L,R16
                LDI R16,5
                OUT TCCR1B,R16



LUPTIMER:
                IN R17,TIFR
                SBRS R17,TOV1
                RJMP LUPTIMER
                LDI R16,4
                OUT TIFR,R16
                POP R17
                POP R16
                SEI
                RET


;-------------------------------------------------------------
DELAY1:     CLI
                PUSH R16
                PUSH R17
                LDI R16,4
                OUT TIMSK,R16
                LDI R16, HIGH(TIMER_1)
                OUT TCNT1H,R16
                LDI R16, LOW(TIMER_1)
                OUT TCNT1L,R16
                LDI R16,5
                OUT TCCR1B,R16

LUPTIMER2:
```

```
                    IN R17,TIFR
                    SBRS R17,TOV1
                    RJMP LUPTIMER2
                    LDI R16,4
                    OUT TIFR,R16
                    POP R17
                    POP R16
                    SEI
                    RET

        .ORG 0X80
        POLA:
        .DB 0X89,0X88,0XC7,0XC0,0XFF,0XFF,0XFF,0XFF ; HALO

        POSISI:
.DB 0XFE,0XFD,0XFB,0XF7,0XEF,0XDF,0XBF,0X7F
```

